

Srishti Gupta

Third Year Undergraduate
Department of Chemical Engineering

 Srishti Gupta |  srishtig21@iitk.ac.in
 Srishtig21 |  +91 9098616032

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology, Kanpur	6.7/10
2021	CBSE(XII)	Sanskar Public School, Gwalior	93.4%
2019	CBSE(X)	Gwalior Glory High School, Gwalior	96.2%

Scholastic Achievements

- Secured an **All India Rank of 5596** in **JEE Advanced 2021** exam, which was attempted by a total of 150K students
- Secured an **All India Rank of 10355** in **JEE Mains 2021**, placing in top 1% among 9.4 lakh candidates who appeared.

Key Projects

Machine Learning in Chemical Engineering *SimuTech Project, IIT Kanpur*

Dec'22-Feb'23

- Gained knowledge of **Linear Regression** and its solutions, including libraries like **numpy, pandas, matplotlib and seaborn**
- Cleaned and preprocessed a dataset of air quality index and implemented linear regression model to predict relative humidity
- Successfully implemented generalized linear and **softmax regression** models and learned about **neural networks**(shallow/deep)
- Developed an **efficient** predictive model using **advanced machine learning algorithm** with help of **library tensorflow**

Vending Machine *Self Project*

Dec'22-Dec'22

- Implemented **full-fledged** vending machine using **Object-Oriented Programming** principles for both buyers and owners
- Enabled **automatic** stock availability update after each purchase, ensuring accurate and optimized **inventory management**
- Added a feature where the owner can **add** new items, while ensuring that buyers **cannot** see or **access** items with zero stock
- Designed and developed a robust **owner-exclusive transaction record system** for comprehensive and reliable tracking

Numerical Methods *Course Project ESO208, Prof Sachchida Nand Tripathi, IIT Kanpur*

Aug'22-Nov'22

- Developed code for **root finding** algorithms including bisection, interpolation, secant, one-point and newton-raphson methods
- Developed code for solving **linear systems** (direct methods) and **matrix eigenvalue** computation using **MATLAB**.
- Successfully utilized **lagrange polynomials** and **cubic splines** for accurate, smooth, and efficient data interpolation
- Gained knowledge about romberg integration, trapezoidal rule and gauss-legendre quadrature for precise 1-D integration

Cyber Physical System *Course Project CS637, Prof Indranil Saha, IIT Kanpur*

Aug'22-Nov'22

- Thourouly analyzed and successfully implemented the reasearch paper authored by Abolfazl Karimi on Automatic Generation of Test-cases of increasing Complexity for Autonomous Vehicles at Intersections that relies on a pail-fail criteria of the test-case
- Introduced an algorithm generating test-case scenarios with increasing complexity, amplifying the likelihood of failure
- Demonstrated the relevance by generating test-cases for 3 different types of intersections with varying scenes and vehicle behaviors
- Executed these test cases to amplify the evaluation of CARLA's autopilot and autopilot-plus-RSS in the CARLA simulator

Technical Skills

Programming Languages	Libraries	Software
C++, Python, C, Java, MATLAB, L ^A T _E X, SQL	Numpy, Pandas, Matplotlib, Scikit-Learn, Seaborn, TensorFlow	Da Vinci Resolve, Autodesk Inventor, Craftware Pro, Git

Relevent Courses

**online, **ongoing*

Fundamentals of Computing	Data Structures and Algorithms,Udemy*	C++ Programming,Udemy*
Introduction to Electronics - (A)	Computational Methods in Engineering	Marketing Management
Embedded & Cyber-Physical Systems**	Linear Algebra and Differential Equations	Fluids Mechanics

Positions of Responsibility

Secretary, Dance Club | IIT Kanpur

Jun'22-May'23

Time Management	- Devoted significant time to practicing and organizing club events while balancing academic commitments
Workshops	- Successfully facilitated workshops by the Ugh crew and the Ace crew ensuring active participation
Participation	- Learned diverse dance choreographies and actively contributed to events like PG Orientation and Footloose
Promotion	- Implemented promotional strategies, such as posters, encouraging batchmates and juniors to participate
Achievement	- Secured 3rd position in the group dance competition(Footloose), held at Thomso'22 fest of IIT Roorkee

Organiser, Videography | Antaragni'22, IIT Kanpur

Dec'22-Mar'23

- Used **Adobe Premiere Pro** and **DaVinci Resolve** to create vibrant fest videos that captured the event's energy and enthusiasm
- Selected and synchronized suitable music tracks that complemented with the visuals, enhancing the overall impact of the footage
- Dedicated **significant** time and utilized skills to accurately portray the spirit and essence of the festival in the final videos
- Produced engaging videos for three events: Theme Revealer, Synchronicity, and Colour Run, showcasing their excitement

- Invited participants from **150+** colleges all around India for **15+** technical competitions, and **10+** Technocruise workshops
- Efficiently organized and oversaw accommodations for **200+** participants, guests, and dignitaries attending the festival
- Successfully addressed and resolved complex concerns and complaints, ensuring prompt and utmost satisfactory outcomes

Extra-Curricular Activities

- Participated in **3** Model United Nations(MUN) conferences earning **1 Special Mention** recognition for diplomacy & negotiation
- Participated enthusiastically in the **9th All India Independence Cup Karate-Do Championship** in the under **8** age category
- Took part in the intellectually stimulating **Dainik Bhaskar Brain Hunt quest**, demonstrating passion for problem-solving
- Enthusiastically competed in **online gaming tournaments** (ValoRush), demonstrating skills in games such as Valorant